PREMIUM BASIC SCHOOL MANAGEMENT SYSTEM

Multi-Tenant Enterprise Software Requirements Specification

Version 2.1 | Adopted Baseline — Approved for Build | Nursery – Kindergarten – Primary – Junior High School

Platform-as-a-Service Edition

Supersedes Version 2.0 (Proposed Baseline) and Version 1.0 (Single-Institution Baseline, July 2026)

Item	Value
Document	PBSMS Multi-Tenant Enterprise SRS
Version	2.1
Status	Adopted Baseline — all Version 2.0 recommendations approved for build; ready for Phase 0 execution
Delivery Basis	Greenfield build — no legacy system exists; there is no data to preserve or migrate
Deployment Model	Single platform, multiple independent school tenants
Country Context	Ghana (default), extensible to other West African markets
Default Currency	GHS (Ghana Cedi) — configurable per tenant
Default Time Zone	Africa/Accra — configurable per tenant
Education Levels	Nursery, Kindergarten, Primary (Basic 1–6), Junior High School (JHS 1–3)
Out of Scope	Senior High School, tertiary education, semesters, credit hours, GPA/CGPA

Build Readiness Status — Version 2.1

This is a greenfield build: no prior system exists, so there is no legacy data, no legacy schema and no working behavior to audit or preserve. Every recommendation carried a 'confirm before treating as final' caveat in Version 2.0. That caveat is now removed. The following are ADOPTED, not proposed, and Phase 0 (Chapter 44) proceeds on this basis:

Decision	Status
Tenancy isolation model — shared database, shared schema, tenant_id + PostgreSQL RLS (Chapter 1)	ADOPTED
Tenant hierarchy — Tenant → School → Campus (Chapter 2)	ADOPTED
Technology stack — NestJS, PostgreSQL 15+, Redis, Next.js, S3-	ADOPTED

Decision	Status
compatible storage (Chapter 29)	
Subscription tiers and metering basis — per-active-student, four tiers (Chapter 5)	ADOPTED as v1 working model; revisit pricing specifically against real numbers once pilot billing data exists (Chapter 45.3) — the mechanism is decided, the exact figures are not yet market-tested
Named vendor integrations — Paystack, Hubtel, MTN MoMo, Telecel Cash, WhatsApp Business API (Chapters 24, 26, 36)	ADOPTED as the integration set; vendor onboarding is a Pre-Phase-0 workstream (Appendix E)
Delivery model — human-led team, phased roadmap, AI-assisted tooling under review (Volume VI)	ADOPTED

A new Appendix E (Pre-Phase-0 Parallel Track) lists everything that is not engineering — legal, vendor onboarding, hiring, pilot recruitment — that should start immediately and run alongside Chapter 44's engineering roadmap rather than after it.

Revision Summary — Version 1.0 → Version 2.0

Version 1.0 specified a single-institution platform. Version 2.0 is a structural rewrite, not an edit: the platform is now specified as a multi-tenant system from the ground up, with every downstream chapter written against that decision. The table below summarizes what changed and why.

Area	Version 1.0	Version 2.0
Tenancy	School is the root aggregate; school_id present "where applicable"	Tenant is the root aggregate; tenant_id mandatory, indexed and enforced at the database layer on every tenant-owned table
Isolation	Not specified	Shared database, shared schema, PostgreSQL Row-Level Security as a second enforcement layer beneath the application
Platform roles	None — only school-level roles existed	Platform Super Admin, Support Engineer, Billing Administrator, Onboarding Specialist added with scoped, audited access
Commercial model	School bills parents only; no platform revenue model	Full subscription, tiering and metering model added (platform bills schools; schools still bill parents)
Requirement IDs	10 IDs total (BR-001–BR-010); traceability claimed but not achievable	Full ID scheme across every functional and non-functional requirement; traceability is now achievable
Non-functional targets	No numeric targets anywhere; deferred to "baseline measurement"	Concrete numeric targets defined for latency, throughput, availability, RTO/RPO (Appendix B)
Compliance	Not addressed	Ghana Data Protection Act, 2012 (Act 843) and GDPR-alignment chapter added, with retention schedule and consent model
Local curriculum fit	Generic numeric/developmental grading only	NaCCA Standards-Based Curriculum, BECE, GES reporting and CSSPS alignment added
Connectivity	Not addressed; "offline" mentioned twice, incidentally	Offline-first attendance and score capture specified as a first-class requirement with a defined sync protocol
Communication	Email-first; SMS "where enabled"; no WhatsApp	WhatsApp Business API specified as the primary channel; SMS and email as defined

Area	Version 1.0	Version 2.0
		fallbacks
Payments	Generic "provider adapter"; mobile money listed as a dropdown value	Named Ghanaian providers (Paystack, Hubtel, MTN MoMo, Telecel Cash) with a concrete settlement and reconciliation model
Accessibility	"Accessible" asserted with no named standard	WCAG 2.1 Level AA named as the binding target, with an automated CI gate
Security pipeline	review → tests → build → staging, no security gate	SAST, dependency scanning, DAST and a pre-GA penetration test added as release gates
Delivery model	55 sequential prompts for an autonomous AI coding agent ("Codex")	Human-led, phased delivery by a named team, using AI coding assistants as a supervised productivity tool, not as the delivery mechanism
Document structure	Identical five-row control table repeated after all 55 chapters (~15–20% of the document)	Universal controls stated once (below); per-chapter text limited to what is actually chapter-specific

Universal Requirement Controls

The following controls apply to every requirement in this document unless a chapter explicitly states an exception. They are stated once here rather than repeated after each chapter.

Control	Requirement
Tenant scope	Every protected action MUST be evaluated within the caller's tenant. Cross-tenant access is denied by default at both the application layer and the database layer (Row-Level Security).
Permissions	Every protected action MUST require server-side authorization and record-scope validation. Client-side hiding of a control is never treated as authorization.
Audit	Material state changes MUST record actor, tenant, time, old value, new value, reason (where applicable) and outcome, in an append-only log.
History	Official records MUST use versioning or reversal. They are never silently overwritten or hard-deleted while retention rules apply.
Reporting	Applicable operations MUST feed role-based dashboards, scheduled reports and follow-up actions within their tenant scope.
Testing	Business rules, permissions, state transitions, calculations, cross-tenant isolation and regressions MUST have automated test coverage before release (see Chapter 45).

How to Read This Document

Requirement identifiers

Every functional requirement carries a stable identifier in the form FR-<MODULE>-<NNN>. Every non-functional requirement carries NFR-<CATEGORY>-<NNN>. Business requirements use BR-<NNN>. Tenancy/platform requirements use TEN-<NNN>. Data-protection requirements use DP-<NNN>. Domain-alignment requirements use DOM-<NNN>. IDs increment in steps of 10 within a module so that requirements can be inserted later without renumbering. The full index is in Appendix A.

Requirement keywords

This document follows RFC 2119-style keyword conventions: MUST/MUST NOT denote a mandatory requirement; SHOULD/SHOULD NOT denote a strong recommendation that may be departed from with a documented reason; MAY denotes an optional capability.

Module prefixes used in this document

Prefix	Module
ADM	Admissions
STU	Student Lifecycle
ACA	Academic Structure
ATT	Attendance
ASM	Assessment
GRA	Grading Engine
RES	Results Management
DOC	Documents & Certification
PRO	Promotion & Completion
FIN	Financial Management
FEE	Fee Structure & Billing
PAY	Payments, Allocation & Receipts
COM	Communication & Notification
ANL	Analytics & Intelligence
OPS	Supporting Operations
SEC	Security
PERF	Performance & Scalability
AVAIL	Availability, Backup & DR
ACC	Accessibility
API	API & Integration

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VOLUME 0

Multi-Tenant Platform Architecture — Chapters 1–6

Chapter 1 — Platform Vision & Tenancy Decision

This chapter states the single most important architectural decision in this specification and the reason every other chapter in this document exists in its current form. PBSMS is a platform that many independent schools use, not software built for one school.

1.1 Product Vision

PBSMS is a multi-tenant, cloud-hosted school management platform serving Nursery-to-JHS basic schools across Ghana. A single deployment of the software serves an unlimited number of independent school customers ("tenants"), each with its own data, users, configuration, branding and billing, while sharing one codebase, one operations team and one set of infrastructure.

1.2 Why Multi-Tenant, and Not a Separate Deployment per School

- Unit economics: basic schools in this market cannot individually fund dedicated infrastructure, so cost must be shared across tenants to reach an affordable per-student price.
- Operability: one platform to patch, monitor, back up and improve is achievable by a small team; one deployment per school is not.
- Speed to market: a new school should be usable within an hour of signup, not after a bespoke deployment.
- Cross-school insight: proprietors who own more than one school, and PBSMS itself for product improvement, need aggregate visibility that separate deployments cannot provide.

1.3 Isolation Model Decision

TEN-001: The platform MUST use a shared database, shared schema architecture with a mandatory tenant_id discriminator on every tenant-owned table, defended by PostgreSQL Row-Level Security (RLS) policies as a database-layer enforcement mechanism independent of application code correctness.

Rationale: at this price point and this stage of company maturity, database-per-tenant or schema-per-tenant introduces operational cost (migration fan-out, connection pool exhaustion, backup complexity) that is not justified by the data volumes involved. RLS closes the single most dangerous failure mode of shared-schema multi-tenancy — a missing WHERE tenant_id = ? clause — by making tenant isolation a property of the database session, not of any individual query.

Decision record

Option considered	Verdict	Reason
Shared DB, shared schema, tenant_id + RLS	Adopted for v1	Lowest operating cost; strong isolation via defence in depth; standard PostgreSQL feature, no bespoke infrastructure
Shared DB, schema-per-tenant	Rejected for v1, revisit at scale	Migration must run once per schema; hundreds of schemas complicate connection pooling and tooling
Database-per-tenant	Reserved as a paid isolation tier	Justified only for large school groups or regulatory demands; offered later as an add-on, not the default

TEN-002: A dedicated-database isolation tier MAY be offered as a premium add-on for large school groups or customers with a specific regulatory requirement, without changing the default tier's architecture.

TEN-003: Every table that stores tenant-owned data MUST include a non-nullable tenant_id column, MUST index tenant_id as the leading column of its primary access-pattern indexes, and MUST express every uniqueness constraint as tenant-scoped (for example UNIQUE(tenant_id, invoice_number), never UNIQUE(invoice_number)).

TEN-004: Every database connection used to serve a tenant request MUST set the RLS session variable for that tenant before any query executes, and connection pooling MUST reset or re-set this variable between reuses so that a pooled connection can never leak one tenant's session context into another tenant's request.

TEN-005: Platform-level tables (subscription plans, feature flags, the tenant registry itself, platform audit logs) are exempt from tenant_id scoping and are instead restricted to platform-role access only.

Chapter 2 — Tenant Domain Model & Hierarchy

2.1 Domain Hierarchy

Platform Operator (PBSMS) → Tenant (billing and isolation boundary; typically one proprietor or one school group) → School (one or more per tenant) → Campus (one or more per school) → Academic Year → Term → Division → Class Level → Class → Subject.

TEN-010: A tenant MUST be able to contain more than one school. This is what enables a proprietor who owns, for example, a Nursery/KG campus and a separate Primary/JHS campus under one business, or a small school group, to operate as one tenant with consolidated billing and one login for shared staff, while still reporting separately per school.

TEN-011: A school MUST belong to exactly one tenant. A student, guardian, staff member and all their records MUST belong to exactly one tenant, transitively through their school.

2.2 Core Tenancy Entities

Entity	Key attributes	Notes
tenant	id (uuid, pk), name, slug, status, plan_id, billing_email, country, default_currency, default_timezone, created_at, suspended_at, trial_ends_at	Root of the hierarchy. status ∈ {trial, active, past_due, suspended, offboarding, closed}
school	id (pk), tenant_id (fk, indexed), name, code, level_scope, address, created_at	One tenant may have many schools
campus	id (pk), tenant_id, school_id, name, address, is_main	Preserves the useful "one school, many campuses" model from v1.0 beneath the tenant layer
tenant_user	id (pk), tenant_id, user_id, is_platform_user (bool, default false)	Distinguishes platform staff acting with tenant access from that tenant's own staff
tenant_subscription	id (pk), tenant_id, plan_id, seats_or_students_licensed, billing_cycle, status, current_period_start/end	Drives Chapter 5

2.3 Cross-Tenant Prohibition

TEN-012: No relationship of any kind MAY exist between records belonging to different tenants, with the sole exception of platform-role access captured under Chapter 3 and explicitly audited under TEN-030.

TEN-013: A user account MUST belong to exactly one tenant, except platform staff accounts, which belong to none and are granted temporary, audited, per-tenant access as described in Chapter 3.

Chapter 3 — Platform Roles & Access

Version 1.0 defined roles only from the perspective of a single school (Proprietor, Headmaster, Accountant, Teacher, Parent, Student, ICT Officer). None of those roles can operate the platform itself. This chapter adds the roles required to run PBSMS as a business.

3.1 Platform Roles

Role	Purpose	Access model
Platform Super Admin	Full platform operation: tenant lifecycle, plan configuration, global monitoring	Break-glass access only; MFA mandatory; every action audited
Support Engineer	Diagnose and resolve tenant-reported issues	Time-boxed, ticket-linked impersonation of a tenant's admin view; explicit tenant consent captured where feasible; full session recording and audit
Billing Administrator	Manage subscriptions, invoicing of tenants, dunning	Access to tenant_subscription and payment records for platform billing only — no access to a tenant's own student/parent financial records
Onboarding Specialist	Set up new tenants, run seed data and initial configuration	Write access limited to a tenant during its onboarding window (TEN-023), auto-revoked on go-live

TEN-020: Platform roles MUST NOT be assignable to any user who is also a member of a tenant, to avoid a support engineer using their own school's account to gain elevated privilege over other tenants.

TEN-021: Impersonation of a tenant user by a Support Engineer MUST require a linked support ticket, MUST be time-boxed to a maximum of 2 hours per grant, MUST be visible to the tenant's own administrators in their audit log in real time, and MUST NOT permit financial reversal, data export, or permission changes without a second platform approver (four-eyes).

TEN-022: All platform-role actions against a tenant MUST be written to both the platform audit log and that tenant's own audit log, so a tenant administrator can always see what platform staff did in their account.

3.2 School-Level Roles (retained from v1.0, unchanged in substance)

- Proprietor / School Director — owns the tenant relationship; sees consolidated reporting across all schools in the tenant.
- School Administrator, Headmaster, Assistant Headmaster
- Academic Coordinator, Examination Officer
- Admission Officer
- Accountant / Bursar
- Class, Subject, Nursery and Kindergarten Teachers
- Librarian, Health Officer, Storekeeper, Transport Officer
- Parent / Guardian, Student
- School ICT Officer — manages that tenant's own configuration; has no visibility into other tenants and no platform role.

Chapter 4 — Tenant Lifecycle

4.1 Lifecycle States

Trial → Onboarding → Active → Past Due → Suspended → Offboarding → Closed. A tenant record MUST always be in exactly one of these states, and every transition MUST be audited with actor, reason and timestamp.

State	Meaning	Data access
Trial	Self-serve or sales-assisted signup, time-limited	Full read/write, capped seat/student count, watermarked report cards
Onboarding	Sales-confirmed conversion, guided setup running	Full read/write; Onboarding Specialist has scoped write access
Active	Paying, in good standing	Full read/write per subscription plan limits
Past Due	Payment failed or overdue, within grace period	Full read/write continues; billing warnings shown to Proprietor/Administrator roles only
Suspended	Grace period expired	Read-only for guardians/students (published results, invoices, receipts remain visible); write access blocked for staff; data is not deleted
Offboarding	Tenant has requested closure or been closed for non-payment beyond policy limit	Full data export made available; countdown to permanent deletion begins
Closed	Retention period elapsed	Data permanently deleted per Chapter 40 retention schedule, except where law requires longer retention

TEN-023: A trial tenant MUST convert to Onboarding only after a plan is selected and a billing method is confirmed (or an approved free-tier exception is recorded).

TEN-024: Suspension for non-payment MUST NOT remove or hide previously published academic results, issued receipts, or historical records from parents and students — a family's access to their own child's history is not leverage for collecting an unrelated invoice.

TEN-025: A tenant MUST be able to request a full, machine-readable export of all of its data at any time while Active, Past Due or Suspended, and MUST automatically receive one at the start of Offboarding.

TEN-026: Permanent deletion at the end of Offboarding MUST follow the retention schedule in Chapter 40 — financial and academic records subject to statutory retention are archived in cold storage rather than deleted outright, with access restricted to compliance and legal requests only.

4.2 Onboarding & Seeding

FR-ONB-010: The system MUST provide a guided onboarding wizard that creates, for a new tenant, a default academic-year structure, a starter grading scale appropriate to the selected divisions, default fee categories, default report-card and receipt templates, and a starter set of KPIs — so that a new school is operable within one working day rather than after weeks of blank-slate configuration.

FR-ONB-020: Onboarding MUST support bulk import of an existing student, guardian and staff roster from CSV/Excel with dry-run validation, duplicate detection and a human approval step before commit, reusing the same import pipeline used for later ad hoc imports (see Chapter 30).

Chapter 5 — Subscription, Billing & Metering

This chapter is new in Version 2.0 and did not exist in Version 1.0 in any form. It specifies how PBSMS, the company, is paid by schools — distinct from Chapter 24, which specifies how a school bills its own parents.

5.1 Plan Tiers (starting point — confirm pricing with the business before build)

Tier	Target customer	Indicative basis	Illustrative limits
Starter	Single small school, under ~150 students	Flat monthly fee	1 school, 1 campus, core modules, email support
Standard	Typical single school	Per-active-student per term	Unlimited campuses for that school, WhatsApp/SMS credits included up to a threshold, priority support
Group	Proprietor with 2+ schools	Per-active-student per term, volume discount	Multi-school roll-up dashboards, consolidated billing, dedicated onboarding
Enterprise / Dedicated	Large groups, regulatory need	Custom	Dedicated-database isolation tier (TEN-002), custom SLA, custom integrations

TEN-030: An active student is defined as a student with at least one active enrolment in the current term. Suspended, withdrawn, graduated and archived students MUST NOT count toward metered usage.

TEN-031: The platform MUST meter active-student counts per tenant per billing period automatically from enrolment data, and MUST make the current count and the resulting projected bill visible to the tenant's Proprietor/Administrator role at any time — no surprise invoices.

5.2 Billing Operations

- FR-BIL-010: Generate a platform invoice per tenant per billing cycle from metered usage and plan terms.
- FR-BIL-020: Support monthly and termly billing cycles, aligned to the Ghanaian three-term academic calendar as an option.
- FR-BIL-030: Accept platform-invoice payment via the same Ghanaian payment rails used for parent billing (Chapter 24), plus bank transfer and card.
- FR-BIL-040: Run configurable dunning on payment failure — reminder, grace period, Past Due state, then Suspended per Chapter 4 — with every step notified to the Proprietor by email and WhatsApp.
- FR-BIL-050: Produce a platform revenue report (MRR, churn, expansion, tenant-level ARPU) for internal PBSMS use only, isolated from all tenant data views.

Chapter 6 — Tenant-Scoped Non-Functional Architecture

Every non-functional concern in Version 1.0 (caching, backup, monitoring, rate limiting) was specified at the level of one school. This chapter restates those concerns with the tenant as the unit of measurement, and is the contract that Chapters 37–38 build on.

6.1 Caching

TEN-040: Every cache key MUST begin with tenant_id. No cache MAY be shared across tenants under any circumstance, including "reference data" caches — a shared curriculum template cache, for example, is namespaced per tenant even if its content is momentarily identical across tenants, so a later per-tenant customization cannot leak.

6.2 Quotas & Noisy-Neighbour Protection

- NFR-PERF-010: Each tenant MUST be subject to a request-rate limit and a background-job fair-share quota so that one tenant's peak load (for example, an entire school entering scores at the Third Term deadline) cannot degrade response times for other tenants.
- NFR-PERF-011: Bulk operations (report-card batch generation, bulk import, mass notification send) MUST run on a dedicated worker pool, separate from interactive request-serving capacity.

6.3 Observability

TEN-041: Every log line, metric and trace MUST carry tenant_id. Dashboards MUST support both a per-tenant view (for support) and an aggregate platform view (for operations), and MUST NOT allow a platform dashboard query to return the content of tenant-owned records — only counts, timings and health indicators.

6.4 Backup & Restore

TEN-042: Backup MUST be taken at the whole-platform level for efficiency, but restore MUST support a per-tenant point-in-time recovery path that does not require restoring or touching any other tenant's current data. This is specified in full in Chapter 39.

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VOLUME I

Enterprise Foundation — Chapters 7–14

Chapter 7 — Introduction & Business Context

7.1 Background

Basic schools typically maintain admissions, attendance, grading, finance and communication as separate, disconnected records. PBSMS replaces this fragmentation with one integrated platform per school, delivered as a shared, multi-tenant service so that the cost of a professional system is achievable for a single proprietor-owned basic school, not only for large institutions.

7.2 Business Context

A school manages people, learning, money, documents, communication, facilities and institutional accountability. PBSMS treats these as connected business domains supported by a common security, data and reporting foundation, delivered multi-tenant so each school's domains stay logically whole while the underlying platform is shared.

7.3 Business Drivers

- Operational efficiency
- Academic quality
- Financial accountability
- Institutional governance
- Parent communication
- Data integrity and retention
- Platform sustainability (Version 2.0 addition): the product must be commercially viable to operate as a multi-tenant service

7.4 In-Scope Education Levels

- Nursery 1 and Nursery 2
- Kindergarten 1 and Kindergarten 2
- Basic 1 through Basic 6
- JHS 1 through JHS 3

7.5 Out of Scope

- Senior High School administration
- University, college or tertiary structures
- Semesters, credit hours, GPA or CGPA
- Lecturers, faculties and degree programmes

7.6 Configurable Regional Defaults

The Ghana defaults (GHS currency, Africa/Accra time zone, three-term calendar) remain the shipped default, but are now explicitly per-tenant configuration values (TEN-010 hierarchy), enabling future expansion to other West African markets without a code change.

Chapter 8 — Business Requirements

8.1 Primary Goal

Provide one integrated, secure, multi-tenant platform for the academic, administrative, operational and financial management of Nursery-to-JHS schools, operable as a sustainable software business.

8.2 Strategic Requirements

- BR-001: Maintain one authoritative source of institutional data per tenant.
- BR-002: Reduce manual workload through safe automation.
- BR-003: Provide timely, role-appropriate decision information.
- BR-004: Support transparent, reconcilable finance operations, both school-to-parent and platform-to-school.
- BR-005: Improve communication with staff and guardians on the channels they actually use.
- BR-006: Enforce accountability through permissions, workflows and audit trails, including platform-staff access to tenant data.
- BR-007: Preserve historical records and published versions indefinitely, subject to the retention schedule.
- BR-008: Support configurable policies without source-code changes, at the tenant level.
- BR-009: Monitor responsibilities using measurable KPIs, at both school and platform level.
- BR-010: Scale horizontally across tenants without architectural redesign (Version 1.0's BR-010 said "scale" with no tenancy model behind it; Volume 0 is that model).
- BR-011 (new): Onboard a new school tenant to a usable state within one working day.
- BR-012 (new): Guarantee that no tenant can access, infer, or receive another tenant's data under any product surface, including support tooling.
- BR-013 (new): Operate the platform profitably at a per-student price point affordable to a proprietor-owned basic school in Ghana.

8.3 Success Measures

Success is measured by: preserved legacy data during any tenant migration, reduced duplicate entry, complete permission and tenant-isolation enforcement, faster processing, reconciled finance at both school and platform level, timely reporting, traceable approvals, successful multi-channel message delivery, and time-to-value for a newly onboarded tenant.

Chapter 9 — Enterprise Design Principles

- 9.1 Tenant Isolation by Default (new): every design decision defaults to isolated; sharing across tenants requires an explicit, reviewed exception.
- 9.2 Single Source of Truth: each business entity has one authoritative owner within its tenant; other modules consume its data through controlled services.
- 9.3 Modular Independence: modules own their business responsibility and evolve without direct manipulation of another module's internal data.
- 9.4 Configuration Before Customization: calendars, class levels, subjects, grading, assessments, fees, reports, branding and workflows are configured per tenant in the application, never hard-coded.
- 9.5 Security by Design: authentication, server-side authorization, tenant and record scope, auditing and sensitive-data controls apply to every protected operation.
- 9.6 Automation with Oversight: routine calculations, reminders, schedules and document generation are automated; admissions, publication, reversals, promotion and elevated access retain approval gates.
- 9.7 Historical Preservation: published academic results, payments, receipts, enrolments and audit records are never silently overwritten or deleted.
- 9.8 Usability and Accessibility: interfaces are responsive, keyboard-usable, WCAG 2.1 AA conformant (Chapter 42) and consistent across roles.
- 9.9 Design for Intermittent Connectivity (new): interfaces assume mobile data constraints and occasional connectivity loss as the normal case for this market, not the exception (Chapter 34).

Chapter 10 — Enterprise System Architecture

10.1 Architectural Objectives

- Multi-tenant isolation as a first-class, testable property
- Modularity and loose coupling
- Scalability and maintainability across a growing tenant base
- Reliable automation
- Secure integrations
- High availability and recovery readiness

10.2 Logical Layers

Layer	Responsibility
Presentation	Dashboards, forms and reports; per-tenant branding applied at render time
Application/API	Requests, validation, tenant-context resolution and transaction coordination
Business domain	Institutional rules and workflows, evaluated within tenant scope
Integration	Internal APIs, events and external connectors, with per-tenant credential scoping
Data access	Repositories, queries, caching and transactions; tenant_id enforced on every query
Database	Master, transactional, historical and audit data, with Row-Level Security policies

10.3 Enterprise Domains

- Platform & Tenancy (new — Volume 0)
- Foundation Services
- Student Lifecycle
- Academic Operations
- Financial Management (school-to-parent and platform-to-tenant)
- Operational Intelligence
- Governance and Compliance

10.4 Communication Rules

Modules communicate through approved services, APIs or events. Direct cross-module table manipulation is prohibited. Cross-tenant communication of any kind is prohibited outside the platform-role paths defined in Chapter 3.

10.5 Event-Driven Processing

Domain events (Admission Approved, Result Published, Student Promoted, Payment Reversed, Tenant Suspended, Subscription Renewed) carry tenant_id as a mandatory field and trigger controlled downstream actions scoped to that tenant.

Chapter 11 — Master Data & Enterprise Data Architecture

11.1 Master Data Domains

Domain	Examples	Owner
Platform master data (new)	Subscription plans, feature flags, integration provider registry	Platform
Institutional master data	School and campuses, academic years and terms, divisions and class levels, classes, rooms, calendars and holidays	Tenant
Academic master data	Subjects, assessment types, grading scales, promotion rules, report templates	Tenant
People & finance reference data	Relationships, nationalities, languages, employment types, fee categories, payment methods, scholarship types	Tenant

11.2 Data Classes

- Platform data (new)
- Master data
- Transactional data
- Reference data
- Generated data
- Audit and security data

11.3 Ownership Model

A business entity has one owner within its tenant and any number of authorized consumers. Ownership controls mutation; permissions control visibility; tenant_id controls the outer boundary that neither ownership nor permission may cross.

11.4 Integrity

- Entity integrity through immutable primary identifiers, scoped by tenant_id.
- Referential integrity through foreign keys, which MUST NOT reference rows in a different tenant.
- Domain integrity through validation and controlled vocabularies.
- Business integrity through workflow and policy checks.

11.5 Lifecycle

Records move through controlled states such as Draft, Submitted, Reviewed, Approved, Published, Locked and Archived. Every transition is authorized, tenant-scoped and audited.

11.6 Security Classification

Class	Examples
Public	Approved announcements
Internal	Timetables and assignments
Confidential	Student, guardian, results and statements
Highly Confidential	Medical, biometric, banking data, IDs and credentials

11.7 Sensitive data (medical details, identification numbers, salaries, banking data, biometric templates, password material and security settings) requires elevated controls, masking and restricted export, and is in scope for the retention and consent rules in Volume V.

Chapter 12 — Enterprise Business Rules Framework

12.1 Tenancy Rules (new)

- No record MAY reference a foreign key belonging to a different tenant.
- No user session MAY carry more than one active tenant context at a time, except platform roles operating under Chapter 3.
- Suspension of a tenant MUST NOT alter or hide historical published data (TEN-024).

12.2 Institutional Rules

- Only one academic year is normally current per school or campus.
- Term dates fall within the academic year.
- Every class belongs to one level and every level to one division.
- Every offered class subject must have a valid configuration before publication.

12.3 Student Rules

- Student and admission identifiers are unique per tenant and never reused.
- A student has no more than one active enrolment per academic year unless an explicit exception is configured.
- Transfers preserve academic, attendance, financial, health and discipline history.
- Student records are archived, not permanently deleted, subject to the retention schedule.

12.4 Attendance, Assessment and Finance Rules (retained from v1.0)

- Only active enrolments appear in registers; one record exists per student, date and configured session.
- Teachers enter scores only for active assignments; submitted and published data is locked until controlled reopening.
- Invoices and receipts have unique numbers, scoped per tenant; payments reference valid obligations and are never deleted; corrections use reversal transactions with approval and reason.

12.5 Reporting Rules

- Reports respect tenant and record scope with least privilege.
- Schedules include weekly, monthly, termly and yearly cycles, plus group-level roll-ups where the tenant contains multiple schools.
- Failed deliveries retry and escalate.

Chapter 13 — Roles, Responsibility, Authority & Scope

Platform roles are specified in Chapter 3. This chapter restates and retains the school-level model from Version 1.0, which was already sound.

13.1 Model

PBSMS separates roles (institutional identity), responsibilities (expected work), permissions (allowed actions) and authority (approval power), all evaluated within a resolved tenant context.

13.2 Permission Actions

- View
- Create
- Edit
- Submit
- Review
- Return
- Approve
- Reject
- Publish
- Reopen
- Reverse
- Export
- Print
- Configure
- Archive
- Restore
- View audit history

13.3 Scope Hierarchy

Tenant → School → Campus → Department → Division → Class level → Class → Subject → Assigned students → Linked children → Record ownership. Tenant is the new outermost scope added in Version 2.0; every scope below it is meaningless without it.

13.4 Delegation & Conflict of Interest

Temporary authority includes delegator, recipient, exact functions, start, end and automatic expiry; delegation is visible and audited. Workflows identify when one user holds overlapping roles and prevent required independent review from being silently bypassed.

Chapter 14 — Operational Intelligence Framework

14.1 Components

- KPI Engine
- Reporting Engine
- Scheduler
- Email/SMS/WhatsApp Service
- Notification Engine
- Action Tracker
- Escalation Engine
- Executive Dashboards
- Group/Multi-School Roll-Up (new)

14.2 KPI Definition

Each KPI carries code, name, responsible role, data source and formula, target and weight, warning and critical thresholds, reporting frequency, supervisor, status, and tenant scope.

14.3 Group Roll-Up (new capability)

FR-ANL-010: Where a tenant contains more than one school, the Proprietor/Director role MUST see a consolidated dashboard comparing collections, attendance, academic performance and outstanding actions across all schools in the tenant, in addition to each school's own dashboard. This directly addresses a capability Version 1.0 could not offer because it had no concept of a tenant above a single school.

14.4 AI Readiness

Future AI features may summarize trends, draft report-card remarks, and flag at-risk students or unusual attendance patterns, but official academic and financial decisions remain with authorized human personnel at all times.

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VOLUME II

Academic & Student Lifecycle — Chapters 15–22

Chapter 15 — Admissions Management

15.1 Purpose

Admissions controls the applicant lifecycle, within one tenant, before any permanent student record is created.

15.2 States

Draft → Submitted → Under Review → Documents Incomplete → Interview/Assessment Scheduled → Waitlisted → Approved / Rejected → Admitted → Cancelled.

15.3 Requirements

- FR-ADM-010: Capture applicant identity, photo, birth/nationality/language/address, previous school, guardian and emergency contacts, medical and learning-support information, supporting documents, and interview/entrance-assessment outcomes.
- FR-ADM-020: Detect potential duplicates using configurable combinations of name, birth date, guardian contact and prior school; matches are surfaced for review, never auto-discarded.
- FR-ADM-030: Perform approval-to-student conversion as one atomic transaction — student, guardian relationships and enrolment are created together or not at all.
- FR-ADM-040: Assign the tenant-scoped admission number at conversion, never before, and never reuse a number.
- FR-ADM-050: Provide applications-received, processing-time, conversion-rate and pending-interview/document KPIs, scoped per school and rolled up per tenant.

Chapter 16 — Student Lifecycle Management

16.1 Central Record

One permanent student profile, scoped to its tenant, connects admissions, guardians, enrolments, attendance, academics, finance, health, discipline, documents, promotion and JHS completion.

16.2 Requirements

- FR-STU-010: Maintain a permanent identifier separate from admission number and from yearly enrolment.
- FR-STU-020: Support multiple guardians per student and multiple students per guardian, with relationship-level flags for primary contact, emergency, pickup, finance and report access.
- FR-STU-030: Support class, campus, inbound and outbound transfers with effective dates, reasons, documents and approval, preserving full history. Inter-tenant transfer (a student leaving one PBSMS-hosted school for another) MUST be modelled as export-then-new-admission, never as a direct cross-tenant record move.
- FR-STU-040: Restrict health information beyond general student data; support discipline investigation, response, guardian contact, follow-up, appeal and positive-behaviour recording.
- FR-STU-050: Record every significant student event chronologically with timestamp, actor, module, description and related evidence (student timeline).
- FR-STU-060: Guardians see only linked learners, and only published or explicitly authorized academic, attendance, financial and communication records, enforced identically whether accessed via web or the parent mobile experience (Chapter 34).

Chapter 17 — Academic Management

17.1 Academic Hierarchy

Tenant → School → Campus → Academic Year → Term → Division → Class Level → Class/Section → Class Subjects → Teacher Assignments.

17.2 Requirements

- FR-ACA-010: Default progression Nursery 1 → Nursery 2 → KG 1 → KG 2 → Basic 1–6 → JHS 1–3 → Completed Basic Education, configurable per tenant.
- FR-ACA-020: Academic years and terms use Planned, Active/Open, Closed and Archived states; score-entry, review, approval and publication windows are separately configurable per term.
- FR-ACA-030: Subjects vary by division and class level; Nursery/KG may use developmental learning areas, Primary/JHS use configurable subject structures, with an optional NaCCA strand/sub-strand mapping (Chapter 41).
- FR-ACA-040: Detect teacher, class and room conflicts before a timetable can be published; provide class, teacher and room views.
- FR-ACA-050: Provide teacher-workload, unassigned-subject, timetable-compliance, academic-progress, curriculum-coverage and promotion-readiness reports.

Chapter 18 — Attendance Management

18.1 Scope

Attendance covers students, teachers and non-teaching staff, feeding communication, promotion and performance monitoring.

18.2 Offline-First Requirement (new — closes a Version 1.0 gap)

FR-ATT-010: Attendance capture on a teacher's device MUST function without an active internet connection, queueing entries locally and synchronizing automatically once connectivity returns, per the offline/sync protocol defined in Chapter 34. This is treated as a release-blocking requirement, not an enhancement, because intermittent connectivity is the normal operating condition for many basic schools in this market, and a register a teacher cannot mark is a register that does not get marked.

FR-ATT-011: Conflicting attendance entries created offline by the same user across devices are resolved last-write-wins by device timestamp; entries created by different users for the same student/date/session are surfaced for manual reconciliation rather than silently overwritten.

18.3 Other Requirements

- FR-ATT-020: Statuses: Present, Absent, Late, Excused, Sick, Suspended, Official Activity.
- FR-ATT-030: Corrections after the submission window require permission and a reason, retaining both original and corrected values.
- FR-ATT-040: Repeated absence, low attendance and missing teacher submissions generate notifications, KPI updates and escalation when unresolved.
- FR-ATT-050: Provide daily registers, weekly trends, monthly summaries, term promotion-eligibility and yearly institutional attendance analysis.

Chapter 19 — Assessment Management

19.1 Requirements

- FR-ASM-010: Support configurable assessment types (class exercise, homework, project, mid-term, end-of-term exam) with weighting that sums to a valid total before publication is permitted.
- FR-ASM-020: Restrict score entry to teachers with an active assignment for that class subject in the current term.
- FR-ASM-030: Enforce configured score bounds; require a status or reason for missing scores.
- FR-ASM-040: Lock submitted and published assessment data until a controlled, audited reopening.
- FR-ASM-050: Support an optional NaCCA competency/sub-strand tagging on assessment items for schools that opt into curriculum-aligned reporting (Chapter 41).

Chapter 20 — Enterprise Grading Engine

20.1 Requirements

- FR-GRA-010: Interpret approved assessment data using a versioned grading policy; the engine never collects scores directly.
- FR-GRA-020: Support developmental grading for Nursery/KG, numerical grading for Primary/JHS, and a competency-based model aligned to NaCCA for schools that opt in, without altering historical outcomes graded under a prior model.
- FR-GRA-030: Each scale item has minimum, maximum, grade, optional point, remark and pass/fail status; ranges MUST NOT overlap or leave unintended gaps.
- FR-GRA-040: Processing pipeline: approved scores → totals → percentage → policy lookup → grade → remark → aggregate calculations → result candidate.
- FR-GRA-050: Ranking is disabled by default where developmentally unsuitable (Nursery/KG) and configurable for Primary/JHS, with competition, dense and none modes and explicit tie rules.
- FR-GRA-060: Undefined scales, incomplete components and invalid weights stop processing and create actionable errors rather than silent defaults.
- FR-GRA-070: Every published outcome retains the exact grading-policy version used, permanently.

Chapter 21 — Results Management Engine

21.1 Workflow

Subject Teacher → Class Teacher → optional Academic Coordinator/Examination Officer → Headmaster → authorized publication.

21.2 Requirements

- FR-RES-010: States: Draft, Submitted, Returned, Corrected, Reviewed, Approved, Published, Locked, Archived.
- FR-RES-020: Publication requires all configured approvals complete, required subjects resolved or formally exempted, term/publication windows valid, an applicable report template available, and no unresolved validation errors.
- FR-RES-030: Published results are never overwritten; reopening requires reason and authority; recalculation creates a new version while preserving the previous publication.
- FR-RES-040: Teachers see assigned records, managers see their authorized scope, guardians see linked learners, students see only their own published results.
- FR-RES-050: Provide student-trend, class-average/pass-rate, subject-performance, division-comparison and promotion-readiness analytics, with an optional BECE-format mock-exam analytics view for JHS (Chapter 41).

Chapter 22 — Document Engine & Promotion / Completion

22.1 Document Engine

- FR-DOC-010: Generate report cards, transcripts, admission letters and completion certificates from approved, versioned data — never from live/unapproved records.
- FR-DOC-020: Every generated official document carries a unique, tenant-scoped reference and a digitally verifiable signature or QR code allowing a third party (e.g., a receiving SHS) to confirm authenticity without contacting the school directly.
- FR-DOC-030: Documents are branded per tenant (Chapter 6, Volume 0) using that tenant's configured letterhead, logo and signatory.

22.2 Promotion & JHS Completion

- FR-PRO-010: Convert approved annual academic evidence into controlled progression decisions: promotion, repetition, conditional promotion, pending, transfer, or completion of Basic Education after JHS 3, without overwriting historical enrolments.
- FR-PRO-020: Apply configurable promotion policies by division and class level; separate system recommendations from authorized human decisions.
- FR-PRO-030: Create the next academic-year enrolment only after approval, closing the prior enrolment rather than mutating it.

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VOLUME III

Finance, Communication & Operations — Chapters 23–28

Chapter 23 — Enterprise Financial Management & Fee Structures

This chapter governs school-to-parent finance. Platform-to-school (subscription) finance is specified in Chapter 5 and is architecturally and data-model separate — a school's parent-facing ledger MUST NOT share tables with the tenant's own platform subscription ledger, even though both may appear in one PBSMS account for the Proprietor.

23.1 Core Principles (retained from v1.0)

- Posted transactions are never silently edited or deleted.
- Every balance is derived from valid transactions.
- Receipt and invoice numbers are unique per tenant.
- Sensitive actions require reasons and, where configured, approval.
- Financial reports reconcile to source records.
- Access is restricted by tenant, school, campus and role.

23.2 Fee Structures

- FR-FEE-010: Fee structures are configurable by academic year, term, campus, division, class level and student category.
- FR-FEE-020: Instalment plans define sequence, amount or percentage and due date; the sum MUST equal the total charge.
- FR-FEE-030: Support full, daily, monthly or percentage proration for late entry or transfer.
- FR-FEE-040: Penalty rules specify trigger, grace period, fixed or percentage amount, cap and frequency, tracked separately from the base charge.

Chapter 24 — Invoicing, Payments, Allocation & Receipts

24.1 Financial Record Model (retained — this was correct in v1.0)

Student obligations are represented by invoices and items. Payments remain independent receipts of value and connect to obligations through allocations. Assistance and adjustments are explicit transactions, enabling a complete explanation of every balance.

24.2 Named Payment Integrations (new — Version 1.0 left this generic)

FR-PAY-010: The platform MUST integrate with the following payment rails at launch, each behind the common provider-adapter contract (Chapter 36):

Provider	Rail	Notes
Paystack Ghana	Card, bank, mobile money aggregation	Primary recommended aggregator for breadth of coverage
Hubtel	Mobile money aggregation, USSD	Strong Ghana mobile money reach
MTN Mobile Money (direct)	Mobile money	Largest single mobile money network in Ghana
Telecel Cash (formerly Vodafone Cash)	Mobile money	Second major network
Manual/offline reference	Cash, bank transfer, cheque	Cashier-entered with reference and reconciliation, for schools/parents without digital rails

FR-PAY-020: PBSMS creates an internal payment reference before any provider call; provider callbacks are validated for provider identity, signature, timestamp, replay, reference, amount, currency and event ID; duplicate delivery is idempotent; provider status maps to a controlled internal state machine; settlement, including provider fees, is reconciled against internal records.

FR-PAY-030: Every payment integration credential is stored per tenant, never shared across tenants, and is rotatable independently per tenant (TEN and SEC chapters).

24.3 Requirements

- FR-PAY-040: Invoice generation, payment capture, verification, allocation, reversal approval, assistance approval, reconciliation, period close and report export are distinguishable, separately permissioned actions; schools MAY enforce maker-checker controls.
- FR-PAY-050: Every posting records actor, source, time, amount, currency, related student, method and reference; multi-record operations use database transactions and idempotency protections.

24.4 Acceptance Criteria

- No duplicate invoice or receipt numbers within a tenant.
- Balances recalculate correctly after each transaction.
- Reversals preserve original records.
- Reports reconcile by period and method, including provider fees.
- Unauthorized cross-campus or cross-tenant access is rejected.
- Retrying a financial request does not double-post.

Chapter 25 — Financial Assistance, Reversals & Reconciliation

25.1 Requirements

- FR-FIN-010: Support scholarships, discounts, waivers and sponsor-funded credit as explicit transaction types, each requiring a reason and, above a configurable threshold, a second approver.
- FR-FIN-020: Reversals preserve the original transaction and create a linked, dated correcting entry; nothing is edited in place.
- FR-FIN-030: Reconciliation proves that system transactions agree with cash, bank, mobile money and payment-provider evidence, including mobile-money settlement reconciliation as its own checked category.
- FR-FIN-040: Provide dashboards for today's collections, unverified transactions, outstanding/overdue balances, collection target vs. actual, unallocated credits, pending reversals/assistance, reconciliation differences and failed deliveries — per school, rolled up per tenant.

Chapter 26 — Communication & Notification Centre

Version 1.0 was email-first with SMS "where enabled" and no WhatsApp at all. In this market that ordering is backwards. Version 2.0 reorders the channel priority to match how Ghanaian parents actually communicate.

26.1 Channel Priority (new)

Priority	Channel	Use	Notes
1	WhatsApp Business API	Primary channel for most notifications: attendance alerts, invoice/receipt, results-published notice, general announcements	Highest open rate for this audience; supports rich media (PDF report cards, receipts) and delivery/read receipts
2	SMS	Short, urgent notices; fallback when WhatsApp is undeliverable or the guardian has no smartphone	Delivery-report handling, sender-ID registration and per-message cost tracking required (FR-COM-030)
3	Email	Detailed statements, formal documents, staff-to-staff communication	Retained as the channel of record for anything requiring a durable, searchable copy
4	In-app / push	Future mobile application notifications	Device registration required; specified as future scope, not v1

26.2 Requirements

- FR-COM-010: Integrate with the WhatsApp Business API (Meta-approved Business Solution Provider) supporting template messages, media attachments and delivery/read status, scoped per tenant with its own registered sender.
- FR-COM-020: Templates carry subject, body, variables, channel, language, sensitivity level and version; a preview validates variable substitution before send.
- FR-COM-030: SMS integration uses a named Ghanaian SMS aggregator with registered sender ID, tracks delivery reports, bounce and failure, and enforces a configurable per-tenant monthly cost threshold with alerting before it is exceeded.
- FR-COM-040: Reports are role-scoped, sent by schedule, optionally attach safe PDFs, use authenticated links for sensitive detail, record delivery state per channel and retry failures with channel fallback (WhatsApp → SMS → email) where the message is urgent and the first channel fails.
- FR-COM-050: Respect communication preferences, guardian relationships, consent (Chapter 40) and cost controls; children's, medical and discipline-related communication use stricter minimization and channel restriction.
- FR-COM-060: Recipients can acknowledge reports, comment, be assigned as owners, set deadlines, attach evidence, and complete or reopen actions; overdue items escalate through configurable supervision levels.

Chapter 27 — Academic & Operational Analytics

This chapter consolidates Version 1.0's academic analytics with the new group/multi-school roll-up capability enabled by Volume 0.

27.1 Requirements

- FR-ANL-020: Provide term-over-term and year-over-year trend analysis at student, class, subject, division and school level.
- FR-ANL-030: Provide a tenant-level roll-up (Chapter 14.3) for proprietors operating more than one school.
- FR-ANL-040: Support configurable, auditable AI-assisted summarization of trends and draft recommendations (Chapter 14.4), always presented as a draft requiring human approval before any action is taken on it.

Chapter 28 — Supporting Operations

28.1 Requirements

- FR-OPS-010: Library — catalogue, circulation (issue/return/renew/fine), member linkage to student/staff records.
- FR-OPS-020: Transport — route and stop management, vehicle and driver records, student-route assignment, optional GPS-based arrival notification integration.
- FR-OPS-030: Health — restricted-access medical records, incident logging, medication administration log, guardian notification for incidents.
- FR-OPS-040: Discipline — incident recording, investigation, response, guardian contact, follow-up, appeal, and positive-behaviour recognition, all tenant- and record-scoped per Chapter 13.
- FR-OPS-050: Inventory — asset register, stock levels, issuance tracking, low-stock alerts feeding the notification engine.

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VOLUME IV

Technical Architecture — Chapters 29–38

Chapter 29 — Technology Stack & Layered Architecture

Version 1.0 deliberately avoided naming a stack, framing PBSMS purely as an upgrade of "the existing Premium Grading System." That system does not exist — PBSMS is confirmed as a greenfield build. This chapter's stack is therefore ADOPTED, not conditional: there is no legacy codebase to inspect, preserve, or migrate onto it, and Chapter 44's Phase 0 proceeds directly to building on this stack rather than auditing a prior system first.

29.1 Recommended Stack

Layer	Recommendation	Rationale
Backend framework	Node.js with NestJS (TypeScript), or Laravel (PHP) as a well-supported alternative	Strong typing, mature ecosystem, easy to hire for in Ghana; NestJS's module system maps cleanly onto Chapter 10's domain boundaries
Database	PostgreSQL 15+	Native Row-Level Security (TEN-001), strong JSON support for flexible configuration, mature and inexpensive to operate
Cache & queues	Redis	Session cache, rate limiting (NFR-PERF-010), and background job queue backing store
Frontend	Next.js (React) as a server-rendered, installable PWA	Enables the offline-first requirement (FR-ATT-010) and fast initial load on constrained mobile data
File storage	S3-compatible object storage	Portable across cloud providers; supports signed URLs for private document access
Background workers	Dedicated worker processes consuming the Redis-backed queue, separate deployable from the API	Required for the bulk-operation isolation in NFR-PERF-011
Hosting	Cloud provider with an Africa- or Europe-proximate region; static assets via CDN	Latency to Ghana-based users and payment-provider webhook endpoints

29.2 Layered Architecture

Presentation → Application/API → Domain → Data Access → Infrastructure, as defined in Chapter 10.2, implemented respectively as: Next.js frontend → NestJS controllers/DTOs → NestJS domain services and policy classes → repository layer using an ORM with mandatory tenant-scoping middleware → PostgreSQL/Redis/S3/mail/queue/monitoring providers.

29.3 Transactions and Idempotency

NFR-API-010: Admission conversion, enrolment, transfer, publication, promotion, invoice generation, payment posting, allocation and reversal MUST execute within explicit database transaction boundaries; retrying any of these operations MUST NOT duplicate records (enforced via idempotency keys on the relevant endpoints).

Chapter 30 — Database Architecture, Multi-Tenant Schema & Migration

30.1 Conventions

- Plural snake_case table names; singular foreign-key fields.
- Every table has an immutable UUID primary key; business numbers (admission no., invoice no.) are separate, tenant-scoped identifiers.
- Date types for dates, timestamps (with time zone, stored UTC) for events, fixed-precision decimal types for money — never floating point.
- Timestamps stored in UTC, displayed using the tenant's configured time zone.

30.2 Mandatory Common Fields (revised — no longer "where applicable")

Field	Rule
tenant_id	NOT NULL on every tenant-owned table; leading column of relevant indexes; enforced by an RLS policy on the table
school_id, campus_id	NOT NULL wherever the entity is school/campus-specific; still validated as belonging to the row's own tenant_id
status, effective dates	Present on every entity with a lifecycle
created_at, created_by, updated_at, updated_by	Present on every table
deleted_at, deleted_by, reason	Present wherever soft deletion is appropriate; hard deletion is prohibited on any table listed in the retention schedule (Chapter 40)

30.3 Row-Level Security

NFR-SEC-010: Every tenant-owned table MUST have RLS enabled with a policy of the form USING (tenant_id = current_setting('app.current_tenant')::uuid), applied to SELECT, INSERT, UPDATE and DELETE. Database roles used by the application MUST NOT have BYPASSRLS. A migration that adds a new tenant-owned table without a corresponding RLS policy MUST fail CI (Chapter 45).

30.4 Integrity

- Foreign keys enforce valid relationships and MUST NOT cross tenant boundaries.
- Unique constraints are tenant-scoped and prevent duplicate admission, enrolment, attendance, score, invoice and receipt records.
- Indexes support tenant scope, foreign keys, academic period, status, date and reference searches.
- Protected records use versioning, reversal or archival rather than deletion.

30.5 Migration

Tenant onboarding import and any bulk migration follow: inventory and profile source data → map fields → back up → test migration in an isolated environment → validate counts and relationships → reconcile academic and finance totals → approve → execute → verify. Every migration documents rollback, and imports are always scoped to a single tenant_id, never run across tenants in one batch.

Chapter 31 — Entity-Relationship Model & Core Table Definitions

31.1 Core Relationship

Tenant → School → Campus, and independently, Student → Enrolment → academic and operational transactions. Student identity remains permanent; enrolment carries class and academic-year context.

31.2 Principal Table Groups

Domain	Examples
Platform (new)	tenants, tenant_subscriptions, plans, feature_flags, platform_audit_logs
Identity	users, roles, permissions, user_roles, tenant_users, sessions
Student	students, guardians, student_guardians, enrolments, transfers
Academic	academic_years, terms, divisions, levels, classes, subjects, teacher_assignments
Assessment	assessment_configurations, assessments, scores, grading_policies, results, result_versions
Finance (school)	fee_structures, invoices, invoice_items, payments, allocations, reversals, reconciliations
Finance (platform)	platform_invoices, platform_payments — structurally isolated from school finance tables
Operations	documents, library_items, transport_routes, health_records, discipline_cases, inventory_items
Governance	workflows, workflow_instances, notifications, notification_deliveries, reports, audit_logs, jobs

31.3 Representative Table Definitions

Illustrative, not exhaustive — the full data dictionary is a build-phase deliverable (Chapter 44) traced back to these entities.

Table	Key columns
students	id (uuid pk), tenant_id, school_id, admission_no, first_name, last_name, dob, gender, status, created_at, updated_at
enrolments	id (pk), tenant_id, student_id (fk→students), academic_year_id, class_id, status, start_date, end_date, closed_reason
invoices	id (pk), tenant_id, school_id, student_id, invoice_no (unique per tenant), term_id, total_amount, currency, status, issued_at
payments	id (pk), tenant_id, school_id, provider, provider_ref (unique per tenant+provider), amount, currency, status, received_at, idempotency_key
allocations	id (pk), tenant_id, payment_id (fk), invoice_item_id (fk), amount, created_at
results	id (pk), tenant_id, student_id, term_id, subject_id, grading_policy_version_id, status, version_no, published_at

31.4 Prohibited Relationships (retained from v1.0, extended)

- No row MAY reference a foreign key belonging to a different tenant_id.
- No score without valid assessment and enrolment.
- No report card without approved result version.
- No allocation without posted payment and invoice.
- No guardian access without active relationship.
- No teacher score entry outside assignment.

- No cross-school or cross-tenant relationship unless explicitly authorized under Chapter 3.

Chapter 32 — API Architecture & Service Contracts

32.1 Requirements

- FR-API-010: APIs expose authorized business capabilities only, never unrestricted database access; internal and external consumers use stable, versioned contracts with server-side validation, permissions, tenant scope and audit.
- FR-API-020: The API is versioned (URI-prefixed, e.g. /v1/) with a published deprecation policy of a minimum 6 months' notice before a version is retired.
- FR-API-030: Every authenticated request resolves exactly one tenant context from the caller's credentials before any handler executes; there is no code path in which a handler can be reached without a resolved tenant (platform-role requests included — see Chapter 3).
- FR-API-040: Rate limits are enforced per tenant and per user (NFR-PERF-010); limit responses use standard 429 semantics with a Retry-After header.
- FR-API-050: A partner/integration API tier (read-mostly, OAuth2 client-credentials, scoped per tenant) is defined for future accounting-system and LMS integrations, even where no integration ships in v1.

Chapter 33 — Application Security, Privacy & Access Control

33.1 Security Model

Least privilege, defence in depth, explicit authorization, secure defaults, data minimization, segregation of duties, accountability and privacy by design — retained from v1.0, now explicitly including tenant isolation as a defence-in-depth layer (database RLS beneath application authorization).

33.2 Authentication & Authorization

- SEC-010: Unique named accounts; no shared operational accounts.
- SEC-020: Adaptive password hashing (Argon2id or bcrypt); rate-limited login and secure reset.
- SEC-030: MFA mandatory for all platform roles (Chapter 3) and for school Proprietor/Administrator roles; optional and encouraged elsewhere.
- SEC-040: Secure session cookies, rotation, timeout and revocation.
- SEC-050: Effective access combines active role, permission, tenant scope, record relationship, record state, workflow authority and delegation; view/edit/approve/publish/reverse/export remain separately grantable.

33.3 Segregation of Duties

- Cashier cannot approve own reversal.
- Score entry is separate from final publication.
- High-privilege role assignment receives extra review.
- Backup creation and production restoration are distinct authorities.
- Platform Support Engineer impersonation cannot perform financial reversal or export without a second approver (TEN-021).

33.4 Application Controls

- Parameterized database queries throughout — no string-built SQL.
- Output escaping and safe rich-text sanitization.
- CSRF protection and secure headers (CSP, HSTS, X-Content-Type-Options).
- File signature validation and private object storage with signed, expiring URLs.
- Protection from open redirects, SSRF, injection and path traversal.
- Secret management outside source control, via a dedicated secrets manager, scoped per environment and per tenant where credentials are tenant-specific (payment provider keys).

33.5 Security Testing Gate (new — closes a Version 1.0 gap)

NFR-SEC-020: The release pipeline (Chapter 45) MUST include, as blocking gates: static application security testing (SAST) on every pull request; dependency and container vulnerability scanning on every build; dynamic application security testing (DAST) against staging before promotion to production; and an independent penetration test at minimum annually and before General Availability. Critical or high findings block release until remediated or formally risk-accepted by a named security owner.

33.6 Data Protection & Monitoring

Security and business audit events are append-only for ordinary users. Children's, medical and discipline data use stricter minimization, retention, export and communication-channel rules (Volume V). Incident response follows: detect, classify, contain, preserve evidence, investigate, recover, notify authorized parties (including tenant administrators and, where required by law, the Data Protection Commission — Chapter 39), correct root cause and monitor.

33.7 Non-Production Data Handling

SEC-060: Production tenant data, in particular any child's personal, medical or biometric data, MUST NOT be copied into a non-production environment. Test and staging environments use a synthetic data generator that produces realistic but entirely fictitious Nursery-to-JHS data at representative volume.

Chapter 34 — Frontend Architecture, UX & Offline/PWA Strategy

34.1 Application Shell

- Header with tenant/school branding, context switcher (for multi-school tenants), search, notifications and user menu.
- Permission-generated collapsible sidebar.
- Main content with breadcrumbs; responsive drawer on small screens.
- Visible active academic year, term and campus.

34.2 Design System

Central design tokens govern colour, typography, spacing and breakpoints, with a tenant-branding layer (logo, primary colour, report letterhead) applied on top without forking the underlying component library.

34.3 Offline & PWA Strategy (new — closes a Version 1.0 gap)

- FR-UX-010: The web application is installable as a Progressive Web App on Android devices, the dominant device class among parents and teachers in this market.
- FR-UX-020: Attendance registers and score entry (the two highest-frequency, most connectivity-sensitive workflows) MUST be usable offline: the required data (roster, assessment structure) is pre-fetched and cached when connectivity is available, entries are queued locally (IndexedDB) when it is not, and a background sync process pushes queued entries once connectivity returns, per the conflict rule in FR-ATT-011.
- FR-UX-030: The interface MUST show a persistent, unambiguous online/offline/syncing indicator, and MUST NOT allow a user to believe an offline-queued action has reached the server until sync confirms it.
- FR-UX-040: Pages MUST be designed for low data usage — no unbounded image loads, paginated lists by default, and a data-saver mode for report viewing.

34.4 Accessibility

See Chapter 42 for the binding WCAG 2.1 AA standard and its CI enforcement. Keyboard access, screen-reader labelling, contrast, focus management, zoom and reduced-motion support are implemented against that standard, not against an unspecified general notion of "accessible."

34.5 State and Security

Loading, success, error, offline, empty and restricted states are visually distinct. The frontend hides unavailable actions for usability, but the server remains the sole authority — no client-side check is ever treated as sufficient authorization.

Chapter 35 — Background Jobs, Reporting & Integration Architecture

35.1 Background Jobs

- FR-JOB-010: Schedules support one-time, daily, weekly, monthly, termly, yearly and event-triggered execution, evaluated in the tenant's configured time zone.
- FR-JOB-020: Bulk-operation jobs (report-card batch generation, mass notification, bulk import) run on the dedicated worker pool (NFR-PERF-011), never on request-serving capacity.
- FR-JOB-030: Every job records tenant_id, start, end, outcome and, on failure, a retry count and dead-letter path with alerting.

35.2 Reporting & Data-Projection Architecture

FR-RPT-010: Reporting queries run against read-optimized projections (materialized views or a reporting replica), never against the primary transactional tables directly for anything beyond a single-record lookup, to protect interactive performance (Chapter 37) under concurrent tenant load.

Chapter 36 — External Integrations & Provider Adapters

36.1 Registry and Credentials

Each integration has type, provider, environment, owner, health, configuration reference and credential reference. Secrets remain outside code, are scoped per tenant where the integration is tenant-specific (payment gateways, WhatsApp sender), and are rotatable and separately managed for sandbox and production.

36.2 Named Integrations at Launch

Category	Provider(s)	Reference
Payments	Paystack Ghana, Hubtel, MTN MoMo, Telecel Cash	Chapter 24.2
Messaging	WhatsApp Business API (Meta BSP), named Ghanaian SMS aggregator, SMTP email provider	Chapter 26.1
Biometric devices (optional)	Standard fingerprint/RFID device integration for attendance	Registered per campus; offline events retain original time and unique IDs
Accounting export (future scope)	QuickBooks, Sage	Named as a defined gap in v1.0 (Section 4.13); scoped here as post-launch

36.3 Reliability

Timeouts, circuit breakers, bounded retries, exception queues and graceful degradation prevent false success and protect core operations. Integration dashboards show health, last success, failures, queue depth, latency and credential expiry, scoped per tenant for tenant-specific integrations and globally for platform integrations.

Chapter 37 — Performance, Scalability, Reliability & Observability

Version 1.0 deferred every number in this chapter to "baseline measurement." Version 2.0 states working targets now; they are to be re-validated, not invented from nothing, once real production telemetry exists, per NFR-PERF-090 below.

37.1 Concrete Performance Targets

Metric	Target	Requirement ID
Interactive page load (p95, 3G-equivalent connection)	≤ 2.5 seconds	NFR-PERF-020
API response time, standard read (p95)	≤ 300 ms	NFR-PERF-021
API response time, standard write (p95)	≤ 500 ms	NFR-PERF-022
Report-card batch generation, 500 students	≤ 5 minutes, as a background job with progress indication	NFR-PERF-023
Concurrent active users per tenant at peak (e.g. Third-Term score entry)	≥ 200, without cross-tenant degradation	NFR-PERF-024
Platform-wide concurrent tenants at peak (Year-1 target)	≥ 150 tenants concurrently active	NFR-PERF-025
Monthly uptime (production)	≥ 99.5%	NFR-AVAIL-010
Recovery Time Objective (RTO)	≤ 4 hours for full platform; ≤ 1 hour for single-tenant restore	NFR-AVAIL-020
Recovery Point Objective (RPO)	≤ 15 minutes	NFR-AVAIL-021
Notification delivery latency (WhatsApp/SMS, non-bulk)	≤ 60 seconds from trigger to provider handoff, p95	NFR-PERF-026

NFR-PERF-090: These targets MUST be re-validated against real production telemetry within the first 90 days of General Availability and revised where the baseline shows they were set incorrectly; they are working numbers deliberately chosen to be testable, not final commitments carved from measurement that does not yet exist.

37.2 Scaling

Stateless application instances, shared session store, database, queues and storage allow horizontal scaling. Web, worker, report, document and integration capacity scale independently, with tenant-aware autoscaling triggers so a small number of very large tenants cannot starve autoscaling headroom from the rest of the platform.

37.3 Database and Cache

Query plans, pagination, indexes, bounded connection pools, short transactions and reporting projections are mandatory. All cache keys are tenant-scoped (TEN-040) and never bypass authorization.

37.4 Concurrency

Optimistic version columns, short locks, unique constraints, atomic updates and idempotency keys prevent lost updates and duplicates for scores, invoices, allocations, receipt numbers, promotion and inventory. NFR-PERF-030: concurrent score entry by two teachers for the same class subject is resolved by field-level optimistic locking with a clear conflict message identifying the other editor, never a silent last-write-wins on the whole record.

37.5 Observability

Structured logs, metrics and traces use correlation IDs and mandatory tenant_id, while excluding secrets. Monitored: latency, errors, database, queues, workers, storage, integrations, backup and business-process health, with per-tenant and platform-wide dashboards (TEN-041).

37.6 Alerts and Testing

Actionable alerts carry severity, owner and escalation path. Load, stress, endurance, failure, concurrency and recovery tests prove capacity and safe failure behaviour against the targets in 37.1, run on a recurring schedule, not only pre-launch.

Chapter 38 — Backup, DR, Testing, Deployment & Release Management

38.1 Backup & Disaster Recovery

- NFR-AVAIL-030: Back up database, uploaded and generated files, configuration, application releases and recovery material on a frequent incremental/log basis, with a daily verified backup, weekly full and retained monthly points.
- NFR-AVAIL-031: Maintain live data plus local/nearline and off-site copies, encrypted, with restricted access, checksums, inventories, retention and immutability where supported.
- NFR-AVAIL-032 (new): Support per-tenant point-in-time restore that does not require restoring or affecting any other tenant's current data — restoring School A after an administrator error must never touch School B.
- NFR-AVAIL-033: Verify archive readability, counts, relationships, files, checksums and migration state on a recurring schedule, not only when a restore is actually needed.
- NFR-AVAIL-034: Document response for hardware failure, corruption, faulty migration, ransomware, account compromise, single-tenant data corruption, facility loss, internet outage and provider outage.

38.2 Testing, Quality Assurance & Traceability

- NFR-QA-010: Automated test coverage MUST meet, at minimum: 80% line coverage on domain/business-logic modules, 100% of permission and tenant-isolation boundary conditions, and end-to-end coverage of every workflow listed as a release-blocking acceptance criterion elsewhere in this document.
- NFR-QA-020: Cross-tenant access attempts are a mandatory, first-class test category — every endpoint, report, export and background job is tested with a second tenant's credentials attempting to read or write the first tenant's data, and the suite fails the build if any such attempt succeeds.
- NFR-QA-030: Migration tests compare counts, identifiers, relationships, scores, results, invoice balances, payments and documents before and after any migration.
- NFR-QA-040: Every requirement ID in this document (Appendix A) MUST map to design, implementation, database change, API, permission rule, automated test and acceptance evidence before the owning chapter is marked complete.
- NFR-QA-050: Defect severities — Critical (security, financial integrity, data loss, cross-tenant leak, total outage), High (major workflow unavailable, official result incorrect), Medium (important impairment with workaround), Low (minor usability/presentation). Critical and High defects block release.

38.3 Deployment, Environment & Release Management

- NFR-DEP-010: Local, development, automated test, staging and production use separate databases, storage, secrets, queues, providers, logs and backups; staging resembles production including representative multi-tenant synthetic data.
- NFR-DEP-020: Pipeline: review → automated tests → SAST/dependency scan (Chapter 33.5) → build → staging deploy → DAST → migration test → acceptance → approval → backup → production deploy → smoke test → monitoring.
- NFR-DEP-030: Migrations affecting tenant-owned tables are additive-first (expand/contract pattern) so that a mid-rollout state never leaves any tenant with a partially-migrated, inconsistent schema.
- NFR-DEP-040: Feature flags support gradual, tenant-by-tenant rollout, retain full permission enforcement, and have a named owner and a removal plan.
- NFR-DEP-050: Rollback triggers include authentication, authorization, academic calculation, financial integrity, cross-tenant isolation, migration and severe performance failures; valid post-release transactions are preserved, and forward-fix migration is preferred over destructive restoration wherever safe.

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VOLUME V

Compliance & Local Market Alignment — Chapters 39–42

Chapter 39 — Ghana Data Protection Act & GDPR Alignment

This chapter did not exist in Version 1.0. PBSMS processes the personal, medical, disciplinary and biometric data of children at scale, hosted in Ghana; this is not an optional chapter.

39.1 Ghana Data Protection Act, 2012 (Act 843)

- DP-010: PBSMS (the company) MUST register as a data controller with Ghana's Data Protection Commission, and MUST support each tenant school in understanding its own controller obligations for the data it enters.
- DP-020: Every category of personal data collected MUST have a documented lawful basis (consent, contract necessity, legal obligation, or legitimate interest), recorded per data category in a data inventory.
- DP-030: The platform MUST provide a data-subject request workflow — access, rectification, and erasure-where-legally-possible — with a defined response SLA (30 days, per Act 843 norms), routed to the correct tenant's administrator with platform oversight.
- DP-040: Any personal data breach MUST be assessed within 24 hours of detection and, where it meets the statutory threshold, reported to the Data Protection Commission and affected tenants without undue delay, per the incident procedure in Chapter 33.6.

39.2 GDPR Alignment

DP-050: For any tenant serving guardians resident in the EU/EEA, or should PBSMS engage EU-based sub-processors, the platform SHOULD operate to GDPR-equivalent standards (lawful basis, data minimization, DPIA for high-risk processing such as biometric attendance, and a Standard Contractual Clauses framework for any cross-border transfer) even though Ghana is the primary jurisdiction.

39.3 Controller/Processor Roles in a Multi-Tenant Platform

DP-060: Each tenant school is the data controller for its own students', guardians' and staff's data. PBSMS is the data processor, acting only on the tenant's instructions as expressed through the product's configuration and the tenant's own staff actions — except for the narrow set of platform-role actions in Chapter 3, which are documented, audited, and limited to what is necessary to operate and support the service.

Chapter 40 — Consent, Retention & Data Subject Rights

40.1 Consent

- DP-070: Communication consent is captured per guardian, per channel (WhatsApp, SMS, email), is versioned, and can be withdrawn at any time without affecting the guardian's ability to view their child's records through the parent portal.
- DP-080: Biometric data collection (Chapter 36.2) requires explicit, separately captured guardian consent distinct from general platform terms, given its Highly Confidential classification (Chapter 11.6).

40.2 Retention Schedule (new — replaces v1.0's undefined "according to policy")

Record type	Retention period	Basis
Core academic record (results, report cards, certificates)	Permanent (archived after graduation, not deleted)	Lifelong reference value; BECE/SHS placement verification
Attendance records	7 years post-graduation, then archived cold storage	Aligned to typical Ghanaian record-keeping norms for schools
Financial transactions (invoices, payments, receipts)	7 years, per standard Ghanaian financial record-keeping practice	Tax and audit requirements
Medical/health records	Duration of enrolment + 3 years, then archived with restricted access	Ongoing care relevance balanced against minimization
Discipline records	Duration of enrolment + 1 year unless a legal matter is pending	Minimization; avoids indefinitely following a minor
Biometric templates	Deleted within 30 days of a student's or staff member's exit from the tenant	Highly Confidential classification; minimization
Audit logs	3 years online, then cold archive for 7 years total	Security investigation and compliance need
Tenant data after account closure (Chapter 4)	90-day recoverable export window, then permanent deletion except statutory-retention categories above	Balances customer recovery against minimization

40.3 Data Subject Rights

DP-090: A guardian or staff member MAY request a copy of their own or their linked child's data, correction of inaccurate data, and — where no retention obligation above applies — erasure. Requests are logged, routed to the tenant, and tracked to the DP-030 SLA.

Chapter 41 — NaCCA, BECE, GES & CSSPS Alignment

This chapter closes the local-curriculum gap identified against Version 1.0, which used a generic numeric/developmental grading model with no link to Ghana's actual curriculum framework.

41.1 NaCCA Standards-Based Curriculum

- DOM-010: Subjects MAY be structured as strands → sub-strands → content standards → indicators, matching the NaCCA Standards-Based Curriculum, as a tenant-level opt-in on top of the generic subject model in Chapter 17.
- DOM-020: Assessment items MAY be tagged to a specific indicator, enabling competency-coverage reporting (which indicators have been assessed, which have not) rather than only a subject-level score.
- DOM-030: Report cards for tenants using the NaCCA model display a competency profile alongside or instead of a raw numeric score, consistent with GES-issued exemplar formats.

41.2 BECE Support

- DOM-040: Support JHS 3 candidate registration with index numbers, matching the format required for BECE registration.
- DOM-050: Support mock-BECE exam management and results in BECE-equivalent format (grade 1–9 scale per subject, aggregate of best six), giving parents and coordinators a realistic preview ahead of the actual examination.
- DOM-060: Provide BECE-readiness analytics: subject-by-subject strength/weakness against historical school performance, at student and class level.

41.3 GES Statutory Reporting

DOM-070: Support GES-standard report-card layout as a selectable template alongside PBSMS's own default, and support the common statutory return formats (enrolment census, attendance returns) that schools are typically asked to produce for GES/District Education Office, as configurable, exportable reports.

41.4 CSSPS Placement Touchpoints

DOM-080: For JHS 3 completers, support recording of CSSPS (Computerized School Selection and Placement System) choices and, where available, placement outcomes, against the student's permanent record — informational recording, not an integration with the CSSPS system itself, which is out of scope for v1.

Chapter 42 — Accessibility & Inclusive Design

DOM-090 / NFR-ACC-010: The platform's staff- and guardian-facing web interface MUST conform to WCAG 2.1 Level AA. This replaces Version 1.0's unqualified use of the word "accessible" with a named, testable standard.

42.1 Requirements

- NFR-ACC-020: Automated accessibility checks (axe-core or equivalent) run in CI on every pull request touching frontend code and block merge on new critical violations.
- NFR-ACC-030: A full manual WCAG 2.1 AA audit, including screen-reader testing, is performed before General Availability and annually thereafter.
- NFR-ACC-040: Critical attendance, score-entry, notification, approval and parent-facing functions are verified usable via keyboard alone and via a screen reader, not merely styled to look compliant.

VOLUME VI

Delivery Strategy — Chapters 43–46

Chapter 43 — Delivery Methodology & Team

Version 1.0's Volume IV specified delivery as 55 sequential prompts executed by an autonomous AI coding agent ("Codex") against an existing codebase, with no team, no estimate, and no named owner for any decision. That approach is replaced here with a conventional, accountable software delivery model. AI coding assistants remain a legitimate and encouraged productivity tool within this model — see 43.3 — but they are a tool a team uses, not the delivery mechanism itself.

43.1 Why This Change

- Accountability: a specification this consequential — children's records, school finances, cross-tenant isolation — needs a named human engineer accountable for each decision, not a prompt-and-hope loop.
- Judgment: several requirements in this document (the tenancy isolation model, the retention schedule, the payment settlement model) require judgment calls, stakeholder sign-off and trade-off decisions that a prompt sequence cannot make.
- Continuity: a real team retains institutional knowledge across the life of the product; a sequence of isolated agent sessions does not.

43.2 Recommended Team Composition (for a v1 build)

Role	Count	Focus
Engineering Lead / Architect	1	Owns Volume 0 and Volume IV decisions; final say on tenancy, security and data model
Backend Engineers	2–3	Domain services, API, integrations, background jobs
Frontend Engineer	1–2	Application shell, offline/PWA, accessibility
QA / Test Engineer	1	Test strategy, the cross-tenant isolation suite (NFR-QA-020), automation
Product / Domain Specialist	1 (can be part-time / founder)	Ghana school-operations knowledge, NaCCA/BECE/GES accuracy, pilot-school liaison
DevOps / Platform Engineer	1 (can be shared/part-time initially)	CI/CD, environments, observability, backup/DR
Compliance/DPO advisor	Part-time / advisory	Act 843 registration and ongoing compliance (Chapter 39)

43.3 AI-Assisted Development

AI coding assistants (general-purpose pair-programming tools) MAY be used by engineers throughout delivery to accelerate scaffolding, test-writing and routine implementation, under the same review, testing and security-gate requirements as any other code (Chapters 33.5, 38.2). No AI tool is granted autonomous commit, deploy, or database-migration authority; every change still passes through the pipeline in Chapter 38.3 and requires human review before merge.

Chapter 44 — Phased Delivery Roadmap

The roadmap below sequences work by dependency, matching this document's structure, and reflects the four-phase plan from the preceding gap analysis, now adopted as the delivery plan of record.

44.1 Phase 0 — Environment & Foundation Setup (Weeks 1–2)

All Phase 0 decisions in Version 2.0 are adopted (Build Readiness Status, front matter). Phase 0 is therefore execution, not deliberation:

- Provision cloud accounts, source control, and the three non-production environments (Chapter 38.3).
- Stand up the repository from the adopted stack (Chapter 29) with the multi-tenant schema and RLS policies from Chapter 30 as the first migration.
- Stand up CI with the security gates from Chapter 33.5 active from the first commit, not retrofitted later.
- File Data Protection Commission registration (Chapter 39) — this has external lead time and should not wait for engineering to reach Volume V.
- Begin the Pre-Phase-0 parallel track (Appendix E) — it runs alongside, not before, this phase.

44.2 Phase 1 — Platform Foundation (Weeks 3–8)

- Volume 0 in full: tenant model, platform roles, RLS-backed database foundation, onboarding flow skeleton.
- Identity, authentication, authorization and audit (Chapter 33).
- CI/CD pipeline with security gates from day one (Chapter 38.3), not retrofitted later.

44.3 Phase 2 — Academic Core (Weeks 9–18)

- Admissions, student lifecycle, academic structure, attendance (including offline/PWA), assessment, grading, results, documents (Chapters 15–22).
- First pilot tenant onboarded onto this slice for early feedback (44.5).

44.4 Phase 3 — Finance, Communication & Operations (Weeks 19–26)

- Fee structures, invoicing, named payment integrations, reconciliation (Chapters 23–25).
- WhatsApp/SMS/email communication centre (Chapter 26).
- Supporting operations (Chapter 28).

44.5 Phase 4 — Hardening, Compliance & Pilot (Weeks 27–34)

- Full NFR validation against Chapter 37 targets under realistic multi-tenant load.
- Penetration test and WCAG 2.1 AA audit (Chapters 33.5, 42).
- 2–3 pilot schools live on the full platform with a structured feedback loop and defined go/no-go criteria (Chapter 45.3).
- NaCCA/BECE/GES alignment features (Chapter 41), if not already folded into Phase 2 by pilot-school demand.

44.6 Phase 5 — General Availability (Week 35+)

- Public/sales-assisted tenant onboarding opened.
- Subscription billing (Chapter 5) fully live.
- Ongoing NFR re-validation against real telemetry (NFR-PERF-090).

Chapter 45 — Quality Gates, Security Review & Pilot Program

45.1 Release Gate (applies to every production release, not only GA)

- All automated tests pass, including the mandatory cross-tenant isolation suite (NFR-QA-020).
- No open Critical or High severity defect (Chapter 38.2).
- SAST/dependency scan clean or risk-accepted by the named security owner (Chapter 33.5).
- NFR targets (Chapter 37.1) validated in staging under representative multi-tenant load for any release touching a high-traffic path.
- Rollback plan documented (Chapter 38.3).

45.2 Pre-GA Security Review

An independent penetration test (Chapter 33.5) and a full WCAG 2.1 AA audit (Chapter 42.1) MUST both be completed, with all Critical/High findings remediated, before General Availability is declared.

45.3 Pilot Program

- Select 2–3 schools reflecting the target range (a small single-campus school and a multi-school proprietor group) as paid or discounted pilot tenants.
- Define explicit go/no-go criteria before pilot start: e.g. zero cross-tenant data incidents, NFR targets met in production, pilot-school Net Promoter Score above an agreed threshold, and no unresolved Critical/High defect for 2 consecutive weeks.
- Run a structured weekly feedback loop with pilot schools throughout Phase 4, feeding directly into the Phase 5 go/no-go decision.

Chapter 46 — Definition of Ready / Definition of Done & Change Control

46.1 Definition of Ready

A backlog item is ready when its requirement ID, scope, dependencies, tenant-scoping implications, data effects, roles, business rules and acceptance tests are known and written down against this document.

46.2 Definition of Done

A backlog item is done only when implementation, migrations (with RLS policy where applicable), permissions, validation, integration, audit events, automated tests (including cross-tenant isolation tests where relevant), documentation and review evidence are complete and traced to its requirement ID in Appendix A.

46.3 Change Control

New or changed requirements are assessed for tenant-isolation impact, dependency impact, migration impact, security impact, cost and schedule impact before entering the roadmap in Chapter 44. A change that weakens tenant isolation (Chapter 1) or removes a compliance control (Volume V) requires sign-off from the Engineering Lead and, for compliance changes, the compliance advisor — no such change may be made silently in the course of unrelated feature work.

APPENDICES

Requirement Index, NFR Summary, Glossary & Change Log

Appendix A — Requirement ID Index (representative sample)

The full traceability matrix (requirement → design → implementation → test → evidence) required by NFR-QA-040 is a living, tool-tracked artifact maintained during delivery. The table below is a representative sample of the ID scheme applied across this document, not the complete list.

ID range	Chapter	Count (this document)
TEN-001 to TEN-042	1–6	19
BR-001 to BR-013	8	13
FR-ADM-010 to 050	15	5
FR-STU-010 to 060	16	6
FR-ATT-010 to 050	18	6
FR-GRA-010 to 070	20	7
FR-PAY-010 to 050	24	5
FR-COM-010 to 060	26	6
NFR-PERF-010 to 090	6, 29, 37	13
NFR-SEC-010 to 060	30, 33	5
NFR-AVAIL-010 to 034	37, 38	8
NFR-QA-010 to 050	38	5
DP-010 to 090	39, 40	9
DOM-010 to 090	41, 42	9

Appendix B — Non-Functional Requirements Summary

All numeric targets in this document, collected in one place for release-gate reference.

Category	Requirement	Target
Performance	Page load (p95, 3G)	≤ 2.5s
Performance	API read (p95)	≤ 300ms
Performance	API write (p95)	≤ 500ms
Performance	Report-card batch (500 students)	≤ 5 min
Scalability	Concurrent users per tenant	≥ 200
Scalability	Concurrent tenants (Year 1)	≥ 150
Availability	Monthly uptime	≥ 99.5%
Availability	RTO (platform / single tenant)	4h / 1h
Availability	RPO	≤ 15 min
Quality	Domain logic test coverage	≥ 80%
Quality	Permission & tenant-isolation coverage	100%
Compliance	Data subject request response	≤ 30 days
Compliance	Breach assessment	≤ 24 hours
Accessibility	Standard	WCAG 2.1 AA

Appendix C — Glossary

Term	Definition
Tenant	A single billing and data-isolation boundary on the platform — typically one proprietor or school group, which may contain one or more schools.
RLS	Row-Level Security — a PostgreSQL feature restricting which rows a database session can see or modify, used here to enforce tenant isolation beneath the application layer.
RTO / RPO	Recovery Time Objective / Recovery Point Objective — maximum acceptable downtime and maximum acceptable data loss in a recovery scenario.
NaCCA	National Council for Curriculum and Assessment — the body responsible for Ghana's Standards-Based Curriculum.
BECE	Basic Education Certificate Examination — the national examination taken at the end of JHS 3.
CSSPS	Computerized School Selection and Placement System — Ghana's system for placing JHS completers into Senior High Schools.
GES	Ghana Education Service.
Act 843	Ghana's Data Protection Act, 2012.
WCAG 2.1 AA	Web Content Accessibility Guidelines, Level AA — the named accessibility conformance target for this platform.

Appendix D — Change Log vs. Version 1.0

See the Revision Summary at the front of this document for the full comparison table. In brief: Version 2.0 adds Volume 0 (multi-tenant architecture, 6 chapters), Volume V (compliance and local-market alignment, 4 chapters) in full; replaces the original Volume IV's AI-agent execution blueprint with Volume VI's human-led delivery strategy; adds a requirement ID scheme and numeric non-functional targets throughout; and removes the repeated five-row control-table boilerplate in favour of the single Universal Requirement Controls section stated once at the front of the document. Version 2.1 confirms the project as a greenfield build and converts every Version 2.0 recommendation into an adopted decision (front matter, Build Readiness Status), simplifies Chapter 44 Phase 0 accordingly, and adds Appendix E.

Appendix E — Pre-Phase-0 Parallel Track

Everything below has external lead time, does not block engineering, and should not wait for Phase 1 to finish. Each item needs a named owner in your organization — the placeholders below are role labels, not assignments.

E.1 Legal & Compliance

Item	Owner	Target	Reference
Register as data controller with Ghana's Data Protection Commission	Compliance advisor / Founder	Week 1–3	Chapter 39, DP-010
Draft Terms of Service and tenant subscription agreement	Legal counsel	Week 1–4	Chapter 5
Draft Data Processing Agreement template for tenant contracts	Legal counsel + Compliance advisor	Week 2–4	DP-060
Confirm data residency / hosting jurisdiction position	Engineering Lead + Legal counsel	Week 1–2	Chapter 29.1, Chapter 39

E.2 Vendor Onboarding

Item	Owner	Target	Reference
WhatsApp Business API approval via a Meta Business Solution Provider	Founder / Ops	Week 1–6 (longest lead time — start first)	Chapter 26.1, FR-COM-010
Paystack Ghana and Hubtel merchant/sandbox accounts	Founder / Ops	Week 1–3	Chapter 24.2
MTN MoMo and Telecel Cash direct integration access	Founder / Ops	Week 2–5	Chapter 24.2
SMS aggregator account and sender-ID registration	Ops	Week 2–4	FR-COM-030
Cloud provider account and Africa/Europe-proximate region selection	Engineering Lead	Week 1	Chapter 29.1

E.3 People

Item	Owner	Target
Confirm or hire Engineering Lead / Architect	Founder	Week 1 — this role should exist before Phase 0 work starts
Hire/assign backend, frontend and QA engineers (Chapter 43.2)	Engineering Lead / Founder	Week 1–4, staggered by Chapter 44 phase needs
Identify a compliance advisor (part-time/advisory is sufficient at this stage)	Founder	Week 1–2

E.4 Pilot Recruitment

Begin identifying 2–3 candidate pilot schools now (Chapter 45.3), even though they will not go live until Phase 4. Recruiting lead time is real, and a proprietor who has already agreed in principle in Week 2 is a very different conversation from a cold approach in Week 27. A good candidate mix: one small single-campus school, and one multi-school proprietor group — the second is what actually exercises the tenant hierarchy (Chapter 2) and the group roll-up (Chapter 14.3) that a single school never touches.

— End of Document —